



The University of Manchester

Light Exposure and Human Behaviour

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Declaration

No portion of the work referred to in this extended research project paper has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Abstract

The study of light exposure and human behaviour has always been an ongoing research topic in Chronobiology, which began in 1729 (Erren [2023]). Chronobiology is a field in biology that studies circadian rhythms which are the 24-hour cycles that occur in organisms, these rhythms are controlled by an internal clock in the organisms with one central clock called the Suprachiasmatic Nucleus (SCN) in the brain that coordinates all internal clocks (Welsh et al. [2010]). This paper investigates the relationships of light exposure history to cognitive performance such as working memory and reaction time, and the sleepiness of the human body by using statistical methods and machine learning models including regression analysis and Random Forest models on experimental data of 60 subjects collected by the University of Manchester during July and August in 2022 and 2023 (Didikoglu et al. [2023]). There is statistical significance between subjective sleepiness and cognitive performance based on the metrics from cognitive tasks psychomotor vigilance task (PVT) (Grant et al. [2017]), N-back (Jaeggi et al. [2010]), and visual search (Wolfe [2016]) each with around 1000 records which matches the associations of sleepiness to the metrics compared to the findings from Gardesevic et al. [2022] with the same set of Linear Mixed models but a different dataset collected by from the same source the Brighter Time smartphone app. Predicting the cognitive metrics with light exposure history information using Random Forest models is unable to capture the underlying relationships between the cognitive metrics and light exposure history due to the lack of samples from the data and the complexity of the light history, yet subject information seems to be more informative than the light history. The results from the analysis of variance of the means of light history information between each distinct score of subjective sleepiness are unreliable due to the lack of requirements for appropriate use of the analysis of variance method.

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Chapter 1

Introduction

Light, alongside other external factors, heavily affects the circadian rhythms in humans, including the sleep-wake cycle when the sleep pattern is disrupted causing sleep deprivation (Borbély [2022]). The disruption of the circadian rhythm can affect mood (Triantafyllou et al. [2019]) and cognitive performances in working memory and reaction time (Lim and Dinges [2010], Killgore [2010]). Scientists aim to propose lighting recommendations to help re-adjust the timing of sleep and improve sleep quality by studying the effect of light on human psychology and physiology (Brown and et al. [2022]). The human body recognises day and night through sensing light characteristics. Melanopsin photoreceptors are one of the light-sensing cells within the eyes that inform the SCN to adjust the circadian rhythms through signalling characteristics of the light sensed by the eye (Alkozi et al. [2017]). This paper analyses data derived from an experiment conducted by the University of Manchester (Didikoglu et al. [2023]) to investigate the impact of light exposure history on humans. Each subject wore a wrist-worn light logger SpectraWear device measuring light exposure. In particular, melanopic equivalent daylight illuminance (Brown [2020]), which measures the impact of the light on the melanopsin equivalent to of daylight's. The Brighter Time smartphone app (Gardesevic et al. [2022]) was used to collect data on cognitive performance and sleepiness. The cognitive performance was evaluated through the cognitive tasks psychomotor vigilance task (PVT) (Grant et al. [2017]), N-back (Jaeggi et al. [2010]), and visual search (Wolfe [2016]), and complete questionnaire through Brighter Time. This paper aims to uncover the relationships between light exposure history and cognitive performance and subjective sleepiness by applying statistical methods and machine learning to the experimental data.

Chapter 2

Literature Review

2.1 Biological Studies

Circadian rhythms are regulated by the interaction of two processes: a homeostatic sleep drive (Process S) and a circadian process (Process C) (Borbély [2022]). A two-process model of sleep regulation provides a foundational framework for understanding how these processes interact to control sleep-wake cycles. Process S increases with the duration of wakefulness and decreases during sleep, while the circadian pacemaker drives Process C, which is synchronized by environmental light-dark cycles.

The melanopsin system is vital in non-visual photoreception (Lucas et al. [2014]). Melanopsins in the intrinsically photosensitive retinal ganglion cells (ipRGCs) signals to the suprachiasmatic nucleus (SCN) of the hypothalamus, and act as the central pacemaker of circadian rhythms. Melanopsin is most sensitive to blue light (around 480 nm) and plays a crucial role in synchronizing the circadian clock to the external light-dark cycle (Vetter et al. [2022]). This synchronization leads to changes in psychological and physiological behaviour, including pupil size (Lazar et al. [2023]), and adjustments in the circadian clock and sleep patterns (Lucas et al. [2014]). Consequently, appropriate lighting can improve work conditions, sleep efficiency, and circadian clock resetting (Brown and et al. [2022]).

Illuminance is a measure of the amount of light incident on a surface area, quantified in lux. It represents how much luminous flux is received per unit area and indicates light intensity (CIE [2000]), and is commonly used to represent light intensity shone on the photoreceptors. Furthermore, melanopsins describe circadian rhythm better than other photoreceptors (Brown [2020]), especially the MEDI calculated based on the spectral power distribution of light, using the D65 standard spectrum as a reference (CIE [1999]), weighted

by the spectral sensitivity of melanopsin (Brown and et al. [2022]).

Spectral sensitivity can be evaluated through post-illumination pupil response (PIPR) metrics. A study by Adhikari et al. [2015], measured the PIPR in 7 participants exposed to light for varying durations (1 second, 10 seconds, and 30 seconds) after being in a dark environment. The results suggest that the plateau and 6-second metrics of the 1-second PIPR are effective in measuring melanopic sensitivity and reflect the response of melanopsins.

Many factors affecting sleepiness are subject-dependent: this includes sleep deprivation (Borbély [2022]), mental disorders (e.g. depression and anxiety) (Krystal [2012]), melanopic sensitivity (Adhikari et al. [2015]), etc., and subjective sleepiness (Shahid et al. [2012], Åkerstedt and Gillberg [1990]). Therefore, the subject identifier is significant in the analysis, and their cognitive performance varies in age (Ding et al. [2019]) and sleepiness (Gardesevic et al. [2022]).

2.2 Measuring Equipment

Gardesevic et al. [2022] investigates the Brighter Time app’s ability to collect data on light exposure’s effect on cognitive performance, based on three tasks performed through the app:

1. **Psychomotor Vigilance Task (PVT):** Measures sustained attention by requiring participants to tap the screen as quickly as possible once a zombie appears.
2. **N-back Task:** Measures working memory by displaying letters one by one on the screen. Participants must tap the screen within a limited time if the current letter matches the letter that appeared N items previously.
3. **Visual Search Task:** Measures search accuracy and efficiency by requiring participants to identify a monkey icon among men icons arranged in different patterns and orientations.

These tasks were carried out in the same order which is PVT, 3-back, 2-back, and lastly Visual Search for consistency. Analysis was performed on experimental data of 69 subjects using the Brighter Time app for a week, fitting linear mixed models on the results of the three tasks individually to investigate the relationship of subjective sleepiness measured by the Karolinska Sleepiness Score (KSS) (Shahid et al. [2012]), light and circadian phase, and daily variation over subjects’ performances.

Casson [2023] presents a wrist-worn sensor node developed for monitoring equivalent daylight illuminance, a measure of light exposure based on how the human eye responds to light. This device is integrated into the Internet of Things (IoT) devices, aiming to provide continuous, real-world light exposure monitoring. The research details the system design, electronic performance, and accuracy of light measurements, comparing them to a calibrated spectral source. The sensor operates over 7 days with a sampling frequency of 30 seconds. It features 10 measurement channels covering a wavelength range of 415–910 nm, balancing accuracy and cost. Findings indicate that the sensor achieves high precision, with a mean absolute error of less than 0.07 log lux and a high correlation between devices (Casson [2023]). This demonstrates the feasibility of accurate light sensing in wearable devices, which could advance both light exposure monitoring and IoT-based lighting control. The data collected by Didikoglu et al. [2023] incorporated this sensor into the wrist-worn light logger SpectraWear to measure α -opic EDI and photopic illuminance.

2.3 Analytical Tools

Circadian rhythms such as MEDI are 24-hour cycles with high complexity and irregularities due to the limitation of measuring devices and subject variability. Fossion et al. [2018], Gao et al. [2023] provides analytical methods to gain insight from circadian rhythms. There are four main categories mentioned each with examples:

1. **Parametric Analysis:** Uses methods that require assumptions about the probability distribution of the rhythm to estimate parameters that characterize the rhythm. In particular, cosinor analysis with a single-component cosinor fit allows easy interpretation of the circadian rhythm (Cornelissen [2014]).
2. **Non-Parametric Analysis:** Opposite to parametric analysis, this approach requires no assumptions about the distribution.
3. **Data-Adaptive Approach:** Methods that adjust their parameters based on the data, such as Empirical Mode Decomposition, which breaks down the signal by modes and analyzes the rhythm in parts.
4. **Nonlinear Approach:** Models the nonlinear nature of the data, such as fractal analysis, which interprets circadian rhythm in terms of fractals, with its degree capturing the rhythm’s complexity and variability.

Breiman [2001] introduces Random Forest, an ensemble learning method that improves prediction accuracy and reduces overfitting by constructing multiple decision trees. It excels in handling complex, high-dimensional data and provides valuable feature importance measures. It is commonly used in biological research (Shao et al. [2018]). However, Strobl et al. [2007] highlight a critical issue: the bias in variable importance measures, particularly in biological data where predictors can vary in scale or categories.

Chapter 3

Methodology

Melanopic activity informs the state of one’s circadian clock and its measurement will be key to uncovering relationships between light exposure and human behaviour. We shall consider subjective sleepiness measured by the Karolinska Sleepiness Scale (KSS) (Shahid et al. [2012], Åkerstedt and Gillberg [1990]) as a score ranging from 1 to 10 going from extremely alert to extremely sleepy; and cognitive performance based on the metrics obtained from completing the cognitive tasks PVT, 2-back and 3-back, and visual search.

3.1 Datasets

There are multiple datasets extracted from the raw data of the experiment, and only datasets helpful for the analyses are mentioned, namely the Brighter Time datasets and SpectraWear dataset. The Brighter Time app data contains dataset with subjects’ demographic including age and gender, mental disorder including ADHD, depression, etc. (See Figure 1 in ‘Additional Figures and Tables.pdf’ in Appendix 6.1); a dataset for Cognitive task metrics throughout the experiment; and a dataset for subjective sleepiness. Subjects were told to perform sessions on the app daily, scoring their subjective sleepiness and completing each cognitive task once each session, up to three times a day. The results in each task are different, with each task performed consecutively in one session, where 2-back is always performed right before 3-back. The number of trials for the subject to respond is fixed for each task, with 28 trials for PVT, 22 for N-back, and 42 for visual search. Each task has different performance metrics, with some available for all the tasks and some being task-specific. All performance metrics are all numeric, their abbreviation and range, if any, are listed below:

Metrics Applicable to All Tasks

- **True Positive Rate (TPR):** The proportion of correct taps out of all correct responses. This provides a measure of sensitivity. Ranges from 0 to 100.
- **False Discovery Rate (FDR):** The proportion of false positives among all the correct identifications. Ranges from 0 to 100.
- **Accuracy (ACC):** The proportion of correctly classified events (both true positives and true negatives) out of all events. This measure is available for all tasks, assessing overall performance correctness. Ranges from 0 to 100.
- **Mean reaction time (meanrt):** The average reaction time across all responses. This measure is recorded for all tasks, giving insight into the general speed of responses.
- **Median reaction time (medianrt):** The median reaction time across all responses.
- **Slowest reaction time (slow10):** The average of the slowest 10% of reaction times, indicating moments of inattention or cognitive delays. This measure is available across all tasks.
- **Fastest reaction time (fast10):** The average of the fastest 10% of reaction times. This provides a measure of peak cognitive performance, and is calculated for all tasks.
- **Inverse Efficiency Score (IES):** Calculated by dividing the mean reaction time by the proportion of correct answers, combining speed and accuracy into a single measure. It is used for all tasks (PVT, N-back, and Visual Search).

Metrics Not Applicable to All Tasks

- **False Positive Rate (FPR):** Used in N-back and Visual Search tasks, FPR is the proportion of false positives out of all actual negatives, assessing the rate of incorrect positive identifications. Ranges from 0 to 100.
- **False Omission Rate (FOR):** Used in N-back and Visual Search tasks. It represents the proportion of false negatives among all actual positives, indicating the likelihood of missing true events. Ranges from 0 to 100.

- **Lapse:** Only exists for PVT. Lapse is defined as the total number of slow reaction times (typically greater than 500 ms), indicating moments of inattention or fatigue. Ranges from 0 to the total number of trials (Grant et al. [2017]).
- **D' (DPRIME):** Only exists for N-back. D' is a measure from signal detection theory that indicates the sensitivity of the subject in distinguishing between signal and noise assessing their working memory (Snodgrass and Corwin [1988]). Also known as D-prime.
- **Slope:** Specific to Visual search. Slope represents the rate of change in performance over time or across difficulty levels in the visual search task (Wolfe [2016]).

Where the metrics that are only applicable to one task are called task-specific metrics.

The SpectraWear data has 1.2 million observations of the subjects' surrounding temperature, α -opic EDI as the EDI for each photoreceptor, photopic illuminance CIE [2000], and 3-dimension acceleration detected through the subject's movement throughout the experiment, measuring once every 30 seconds. The subject is allowed to remove SpectraWear during the experiment leading to missing measures throughout the observations (Didikoglu et al. [2023]).

Most information of the data is excessive to the goal of this paper. Hence data pre-processing or analysis is required to reduce the complexity. Data manipulation and analysis are performed in RStudio. For more details on the datasets and variables, see Appendix 6.1.

3.2 Data Manipulation and Transformation

3.2.1 Visualization

The distributions of cognitive task metrics, KSS, and light measures across the subjects can be observed through boxplots, scatterplots, and histograms.

3.2.2 Timestamp alignment

All subjects participated in the experiment on different dates, so the timestamps of their observations are not aligned with each other. Moreover, the observations of each variable of interest including KSS and cognitive task

metrics from the Brighter Time data, and MEDI from the SpectraWear data were measured at different frequencies or non-concurrent. For the paper’s objectives, the past 24-hour MEDI observations of each cognitive task metric timestamp were aggregated into hourly means and then added to their respective cognitive task metric timestamps in the Brighter Time data. This aggregation lowers the complexity for future analysis with minimum information loss through averaging observations hourly.

3.2.3 Cosinor analysis

The single-component cosinor model (Cornelissen [2014]) can be expressed as:

$$Y(t) = M + A \cos\left(\frac{2\pi t}{\tau} + \phi\right) + \epsilon(t), \quad (3.1)$$

where:

- $Y(t)$ represents the observed value at time t .
- M is the Midline Statistic of Rhythm (MESOR), which is the average level of the rhythm.
- A is the amplitude, indicating the extent of variation within the cycle.
- ϕ is the acrophase, which reflects the phase or timing of the peak of the rhythm.
- $\frac{2\pi t}{\tau}$ represents the periodic component of the model with a period τ (24 hours).
- $\epsilon(t)$ is the error term, capturing any deviations from the expected rhythm due to unexplained variability.

This formulation describes how the observed data $Y(t)$ is modelled as a combination of a periodic cosine function representing the circadian rhythm and an error term $\epsilon(t)$ that accounts for random fluctuations or measurement noise. It is a common practice to use fit a cosinor model to summarise and capture the fluctuation and time phase of circadian rest-activity patterns (Fossion et al. [2018], Gao et al. [2023]).

Using the `cosinor` R package, a single-component cosinor model is fitted on each cognitive observation’s MEDI history in the past 24 hours and the parameters MESOR, amplitude, and acrophase were stored as new variables and added to the Brighter Time data with the hourly mean from Section 3.2.2.

3.2.4 $\log_{10}$ -transformation

$\log_{10}$ -transformation reduces the noises and the influence of outliers, which will be used on MEDI measures. It helps to satisfy normality assumptions for ANOVA test (Montgomery [2019]).

3.2.5 Missing Values

Missing values are handled differently depending on the variable. All the imputations performed are stated in 'Data Attributes and Imputation.pdf' in Appendix 6.1.

3.3 Statistical Methods

3.3.1 Correlation Analysis

Correlation analysis assesses the correlation between variables to investigate the strength of their linear relationship. The Pearson correlation coefficient τ is used throughout the paper Scheer et al. [2013], see Appendix 6.2. This analysis also helps avoid multicollinearity and reduce complexity for other analyses by selecting a set of variables that are weakly correlated or linearly independent, it reduces redundant analysis and simplifies the interpretation.

3.3.2 Analysis of Variance (ANOVA)

Analysis of Variance (ANOVA) is a statistical method used to compare means across multiple groups to determine if there is a statistically significant difference between them through hypothesis testing (Montgomery [2019]). Generally, ANOVA involves a categorical variable $\mathbf{G}$ with k scores, and a continuous variable $\mathbf{Y}$ with N observations. Each observation in $\mathbf{Y}$ is in one of the scores from $\mathbf{G}$. In this paper, one-way ANOVA will be applied on KSS $\mathbf{K}$ and numeric variables of light history (See Appendix 6.3).

Furthermore, the following assumptions must be met for an adequate model fit:

- Normality: For each group of score i , the observations in $\mathbf{Y}_i$ is normally distributed;
- Homogeneity of variance: The variance of every $\mathbf{Y}_i$ is identical.

The difference in group means in $\mathbf{K}$ with $\mathbf{Y}$ can be assessed through an F-test with the null hypothesis H_0 being all group means are equal, and

the alternative hypothesis H_A with at least one group mean differs from the rest (Snedecor and Cochran [1989]), which is evaluated with the F -statistic (See Appendix 6.4). F -statistic lower than 0.05 presents sufficient evidence to reject the null hypothesis and accept the alternative hypothesis implying significant difference between the groups.

One-way ANOVA applies to the score KSS and the numeric variables of light history. With the assumptions satisfied, the impact of KSS on light history can be effectively assessed. However, the assumptions are difficult to meet in real data, but statistical tests can be performed to find significant evidence to meet the assumptions.

- Shapiro-Wilk test (Shahid et al. [2012]) checks for normality, if the p-value is greater than the chosen significant level then the data is not normally distributed;
- Levene's test (Levene [1960]) checks for homogeneity of variance, if the p-value is greater than the chosen significant level, then the groups do not have the same variance (R package `lsr`).

The assumption check process begins with the numeric variable of light history partitioned by KSS. Then performs the Shapiro-Wilk test to each group of KSS individually and selects the normal groups. Finally, apply the Levene's test to check the homogeneity of variance with the selected normally distributed groups.

3.4 Machine Learning

Machine learning models can find complex relationships between light history and cognitive task performance. Several models were trained by features related to the light history with each model predicting different cognitive performance metrics (mentioned in 4.3). This requires substantial computational power. Since interpretability is key in understanding the complex relationship resulting from a fitted model, the following is used for the analysis:

3.4.1 Random Forest

The Random Forest model is a supervised learning method used for classification or regression tasks (Breiman [2001]). This model is constructed from an ensemble of decision trees, each trained on a random subset of the training data, where each tree considers a random set of features when determining

the best splits at its nodes. The overall prediction from the Random Forest is obtained by aggregating the predictions from all individual trees, typically by averaging their outputs. This ensemble approach provides robustness against overfitting and ensures that the model captures the underlying relationships in the data more effectively than a single decision tree would. An important aspect of using Random Forest in regression tasks is the ability to determine the importance of each feature in predicting the target variable and interpreting the influence of the feature on the target variable.

In this study, using the `randomForest` R package, Random Forest regressors with 100 trees, one-third of the number of variables considered at each split, and a maximum of 5 nodes per tree were trained on cognitive task metrics as the target variables. Features were selected from MEDI history hourly means, statistical summaries, cosinor parameters, the task variable, and the subject variable. To evaluate the contribution of each feature to the model’s predictions, the Mean Decrease in Impurity (MDI) method is used. MDI measures feature importance by assessing the total reduction in variance, so how much each feature reduces the uncertainty in the target variable across all trees in the Forest. Specifically, MDI calculates the cumulative reduction in variance at all nodes where the feature is used to split the data, summing this reduction across all trees.

The reason for using MDI as a measure of feature importance in this context is multifaceted. First, MDI directly aligns with the goal of regression, which is to minimize the variance in the predicted cognitive task metrics (Hastie et al. [2009]). By focusing on the total reduction in variance, MDI provides a clear and interpretable measure of how much each light history feature contributes to predicting cognitive performance. This is crucial in understanding the impact of different aspects of light exposure on cognitive function and is key to our research objectives.

The use of MDI as a feature importance metric in this Random Forest regression model is particularly appropriate for the study of cognitive task metrics as it provides a direct and interpretable measure of how light history influences cognitive performance, while also ensuring computational efficiency and stability (Hastie et al. [2009]). However, the interpretability of the relationships with feature importance decreases with an increased number of features which cannot explain all their interactions within the Random Forest model.

The effectiveness of the Random Forest model and the reliability of the MDI feature importance depends on the model’s ability to generalise well to unseen data (Strobl et al. [2007], Hastie et al. [2009]). To ensure that the Random Forest model used in this study does not overfit to training data,

a stratified train-test split is performed, where 80% of the data is used for training and 20% for testing, with these sets having an even distribution of the distinct target values. The model’s performance is then evaluated on both the train and test sets using metrics such as the Root Mean Squared Error (RMSE) and R-squared (R^2), see Appendix 6.6. The RMSE measures the average magnitude of the prediction errors, with lower values indicating better performance. R^2 quantifies the proportion of variance in the target variable that is explained by the features, with a higher value closer to 1 indicating a stronger explanatory power.

By comparing the RMSE and R^2 between the training and test sets, it can be determined whether the model is overfitting. If the model performs better on the training set with significantly lower RMSE and higher R^2 than on the test set, it may be overfitting, capturing noise in the training data rather than general patterns (Chicco et al. [2021]). Conversely, if the model performs poorly on both sets, it may be underfitting, indicating that it is too simplistic to capture the complexities of the data.

Further analysis, Random Forest models will be trained with all observations from all tasks on the shared metrics across tasks, to assess whether the MEDI history is informative to predicting the metrics, in order words, associative. Then the same set of features will be used to train Random Forest models on observations split by task on all the performance metrics including the task-specific metrics.

3.4.2 Linear Mixed Model (LMM)

A Linear Mixed Model (LMM) extends the simple linear regression model by incorporating both fixed and random effects Pinheiro and Bates [2000]. This approach is particularly suitable for hierarchical or longitudinal data, where observations may be nested within groups or subjects, and it allows for the identification of relationships at different levels of variability.

The general form of the LMM used in this study is expressed as:

$$Y_{ij} = X_{ij}\beta + Z_{ij}b_i + \epsilon_{ij}, \quad (3.2)$$

where Y_{ij} represents the response variable for the i -th group (or individual) at the j -th observation. In this model:

- X_{ij} denotes the fixed effect predictors associated with the response. Fixed effects are parameters assumed to have consistent effects across all observations, representing the overall impact of specific predictors on the response variable.

- β is the vector of fixed effect coefficients, capturing the effect size of each fixed predictor on the response variable.
- Z_{ij} represents the random effect predictors. Random effects account for variability in the response variable that is not explained by the fixed effects and varies between different groups or individuals. They model the subject-specific or group-specific deviations from the overall fixed effect.
- b_i denotes the random effects associated with the i -th group or individual. These effects capture the variability in the response attributable to the group or individual level.
- ϵ_{ij} is the residual error term, reflecting any additional variability in the response variable not explained by either the fixed or random effects.

In this analysis, the `lme4` R package was used to fit the LMMs, while the `lmerTest` R package was employed to perform Wald tests [Wald, 1943] on the fitted LMM (See Appendix 6.5 for Wald test). A small p-value (typically less than a significance level α , e.g., 0.05) from the Wald test indicates that the observed correlation is unlikely under the null hypothesis, suggesting that the corresponding fixed effect is statistically significant.

This approach ensures that the significance testing of fixed effects in the LMM accounts for the hierarchical structure of the data, providing more reliable inference compared to methods that ignore this structure.

For each task, three different LMMs were fitted on metrics such as median reaction time, accuracy or hit rate, false alarm rate (FAR), inverse efficiency score, and task-specific metrics (e.g., lapse, D-prime, slope), depending on the task:

- **Model 1:** Includes KSS as a fixed effect to observe the impact of subjective sleepiness on the metrics.
- **Model 2:** Incorporates time awake as the time of day in hours, and ambient light as $\log_{10}$ -photopic illuminance measured from the phone screen, and their interaction as fixed effects to study the effect of circadian phase on the metrics.
- **Model 3:** Utilizes cosine and sine of $\frac{2\pi h}{24}$ (where h is time in hours, e.g., 23:30 is 23.5h) as fixed effects to investigate daily variation effects on the metrics.

All models include the subject variable as a random effect, acknowledging that sleepiness, task performance time are subject-specific and can vary randomly. See Figure 2 and Figure 3 from 'Additional Figures and Tables.pdf' in Appendix 6.1.

The results of the Wald test from these models were compared to those reported by Gardesevic et al. [2022], who applied similar LMMs to a different dataset. This comparison aims to validate the robustness and consistency of the findings and the Brighter Time smartphone app, ensuring that the results in this study align with previously established patterns and thus support the reliability of the conclusions drawn from the new data.

Chapter 4

Results

4.1 Exploratory Data Analysis

From the Brighter Time data, there are 5415 cognitive task observations from the 60 subjects, which are evenly distributed between the tasks PVT, 2-back, 3-back, and Visual Search.

Photoreceptors exist in the retina closely, therefore, the amount of light they sense scales unanimously. This is reflected in Figure 4.1 as it shows all illuminance and EDI being strongly positively correlated. Therefore using one of the EDI from the SpectraWear data can reduce the complexity of analyses. Hence, the MEDI measure is used which is more closely related to circadian regulation (Brown and et al. [2022]).

Every subject presents a circadian MEDI pattern like a light-dark cycle where a long period of minimum illuminance around 0.1 lux presents during sleep, whilst increases gradually as the sun rises till it reaches a higher lux after wake and fluctuates throughout the day until sleep. All subjects work during the daytime with some having occasional night shifts. See more in the 'figure' in the GitHub repository in the Appendix 6.1.

4.2 Data Preprocessing

As mentioned in Section 4.1, there exist missing values and different timestamps in the two data, which need to be treated to extract the correct time information. The subject's MEDI observations in the past 24 hours from when the specific cognitive task was initiated were extracted as history. Since subjects were allowed to remove the SpectraWear, this resulted in missing observation over a long period. Consequently, variables were formed to summarize each subject's MEDI history:

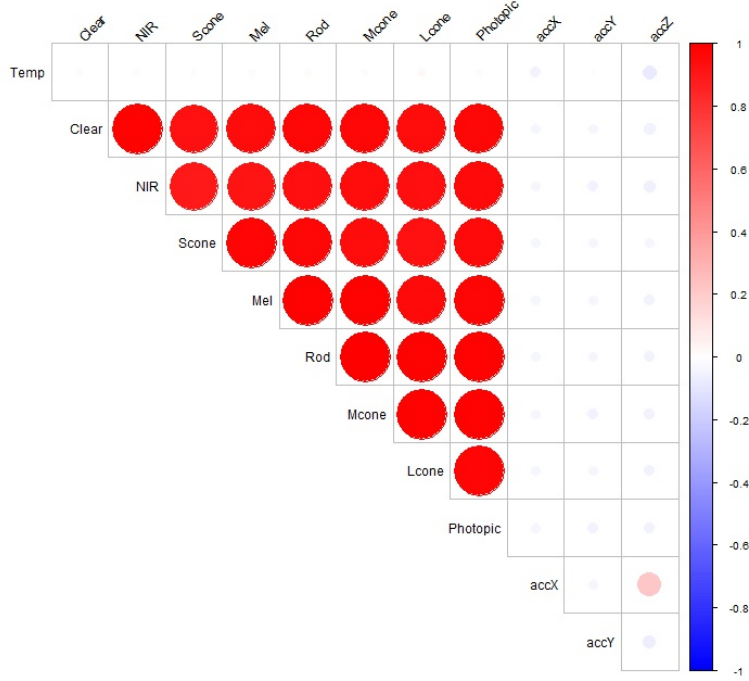


Figure 4.1: The heatmap of the Pearson correlation coefficients between surrounding temperature, all photoreceptors’ EDI, and dimensional accelerations. Each pair of illuminance measure has a coefficient close to 1, implying a fixed difference between each illuminance, presenting strong linear relationships. Acceleration X and Z are weakly correlated; other measures present no correlation.

- The $\log_{10}$ -transformed hourly mean MEDI from each of the 24 hours before.
- Summary statistics of all observations including the mean, minimum, maximum, etc., calculated without 0.1 lux observations to capture the MEDI history during wake.
- The parameters mesor, amplitude, and acrophase of the cosinor regression fitted to the $\log_{10}$ -transformed history in a timely order to capture the circadian patterns.

These variables are added to each cognitive task performance dataset according to the initiation time stamp, and the subjects.

The task observations of subjects with observations less than 8 were removed, allowing comparison across different KSS and cognitive metrics levels appropriately with sufficient samples; also, removing observations consisting of a

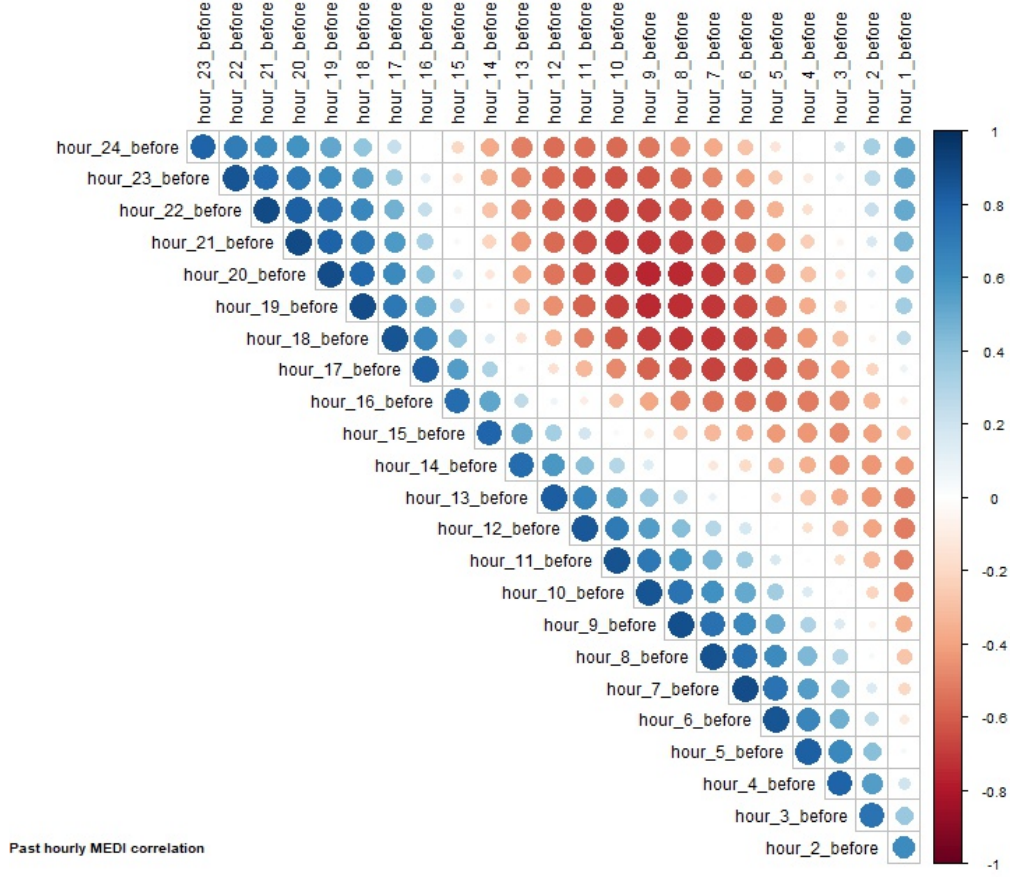


Figure 4.2: Heatmap of the Pearson correlation coefficient between the $\log_{10}$ -transformed hourly MEDI means. It shows the temporal dependency in these means, with strong positive correlations between consecutive hours, and strong negative correlations between means that are 12 hours apart. The strength of the correlation reduces for the more recent observations.

reaction time less than 100ms which is the current limit (Lipps et al. [2011]). The working dataset formed contained 4118 observations and 59 variables. Each task consists of approximately 1000 observations with available MEDI history from 59 subjects, these sessions are not evenly distributed across said subjects.

Some variables were further excluded from analysis due to high correlation. See Figure 4 in 'Additional Figures and Tables.pdf' in Appendix 6.1.

4.3 Analysis

4.3.1 Subject-based cognitive performance

Cognitive performance is subject-dependent, regarding subjective sleepiness. As mentioned in Section 3.4.2, the 3 LMMs will be fitted on 5 performance metrics on the working data. Their Wald test results on each term of the model are compared to the results in Gardesevic et al. [2022], where a p-value less than 0.01 significance level indicates significant evidence for an existing linear association between the fixed effect and target of the LMMs. Different populations and experiment designs have different variabilities, and by comparing the data from Gardesevic et al. [2022] to the data in this paper, there may be a different conclusion drawn to the associations to cognitive performance. The working data in Gardesevic et al. [2022] involves 69 subjects performing sessions of tasks in the same order during the year 2021, with 1305 observations of PVT, 586 observations of 2-back after completing 3-back, and 1215 observations of visual search.

The working data has 1012 observations from PVT, 1032 from 2-back, and 1007 from visual search. 2-back observations were performed before 3-back. Unlike Gardesevic et al. [2022], the 2-back was followed after 3-back.

Table 4.1 indicates that subjective sleepiness (KSS) significantly impacts cognitive performance across tasks:

In PVT, where the median reaction time, lapse, false alarm rate and IES have p-values below 0.01. This suggests that as sleepiness increases, leading to reaction times slow down (5.37 per KSS), lapses increase (7.60 per KSS), depending on the sign of the estimated coefficients. These findings are consistent with Gardesevic et al. [2022], which also reported significant effects of KSS on PVT’s median reaction time and lapse. However, unlike Gardesevic et al. [2022], the study found a significant effect on the false alarm rate.

In 2-back, median reaction time and IES are significant. There are weaker associations between KSS and D’, accuracy, and false alarm rate, with p-values below 0.05. This suggests a potential but weak influence of sleepiness on working memory, contrasting with Gardesevic et al. [2022], where no significant impact of KSS was found on these metrics except IES.

In visual search, median reaction and IES are significant. In contrast, Gardesevic et al. [2022] shows an increase in false alarm rate and a decrease in accuracy as sleepiness increases but agrees with other metrics.

Table 4.2 reveals minimal effects of time awake and ambient light on most

performance metrics. In PVT, an increase in ambient light is associated with reduced median reaction time, likely due to increased alertness, which aligns with KSS findings. In visual search, an increase in time awake is associated with a decrease in accuracy in visual search. In contrast, Gardesevic et al. [2022] presents significance in PVT and visual search such as the positive correlations of median reaction time and slope to time awake, but no impact from the circadian phase to 2-back performance was found.

Table 4.3 shows the cosine term of daily variation significantly decreases IES and median reaction time across tasks, particularly around midday where $h = 12$ with maximum cosine. PVT exhibits strong effects, with p-values well below 0.01, while 2-back and visual search are less influenced. Most results agree with Gardesevic et al. [2022], but not on visual search, suggesting that these tasks might respond differently to circadian rhythms with significance in the sine term and not cosine for median reaction time and IES, with estimated coefficients greater than any estimated coefficients in Table 4.3.

4.3.2 Subjective Sleepiness and Light History

This section investigates the relationship between MEDI history and KSS scores, aiming to determine if there are significant differences in MEDI variables from Section 4.1 across varying levels of subjective sleepiness. ANOVA was employed. However, the assumptions required for ANOVA were not met. Neither normality nor homogeneity of variance was satisfied across all groups, as indicated by the Shapiro-Wilk and Levene’s tests. Consequently, the results from the ANOVA are unreliable.

Combining KSS scores into larger groups did not solve the problem, as the MEDI means distribution remained largely unchanged across the scores. Figure 4.3 illustrates those distributions for KSS scores ranging from 2 to 5 exhibit peaks around 2 to 3 $\log_{10}$ lux with comparable ranges, indicating no significant differences in means among these groups. Distributions for other KSS scores show irregular patterns due to limited data points. This suggests that there may be insufficient evidence to link MEDI history with subjective sleepiness in the dataset.

4.3.3 Effect of Light History to Cognitive Performances

This section investigates the impact of MEDI history on cognitive performance, multiple Random Forest models were trained on each cognitive performance metric using different sets of features, as described in Section 3.4.

Table 4.1: Linear Mixed Models fitted on median reaction time, accuracy, false alarm rate, IES, and a task-specific metric (lapse, D', slope) for each task with KSS as fixed effect and the Wald test results. The p-value is bold if it is less than 0.01 significance level.

(a) PVT

Target		medianrt	lapse	ACC	FAR	ies
(Intercept)		396.98	4.83	94.51	2.22	453.36
kss	Coef	5.37	0.27	-0.20	0.23	7.60
	Std Error	0.933	0.076	0.124	0.077	1.873
	p-value	1.17×10^{-8}	4.11×10^{-4}	1.03×10^{-1}	3.21×10^{-3}	5.41×10^{-5}

(b) 2-back

Target		medianrt	D'	ACC	FAR	ies
(Intercept)		575.31	2.89	95.24	3.51	649.48
kss	Coef	12.39	-0.02	-0.29	0.19	18.42
	Std Error	2.766	0.010	0.114	0.090	3.426
	p-value	8.32×10^{-6}	2.26×10^{-2}	1.06×10^{-2}	3.30×10^{-2}	9.51×10^{-8}

(c) VisualSearch

Target		medianrt	slope	ACC	FAR	ies
(Intercept)		2355.70	653.48	89.19	0.97	2739.34
kss	Coef	42.03	11.64	-0.27	0.08	59.76
	Std Error	9.375	8.841	0.128	0.048	11.218
	p-value	8.24×10^{-6}	1.88×10^{-1}	3.68×10^{-2}	9.04×10^{-2}	1.24×10^{-7}

The performance and generalization of these models were assessed by comparing the Root Mean Square Error (RMSE) and R-squared (R^2) scores of the train and test sets. A stratified train-test split is used with an 80:20 ratio, ensuring the target metric is used for stratification.

The training process involved two approaches: models were either trained using all observations across tasks, with the task variable included as a feature, or trained individually on each task, focusing on both shared and task-specific metrics. Following this, the feature importances of the well-performing Random Forest models were investigated.

There are 2 different sets of features used for each training process. The first set contains the MEDI hourly means up to 11 hours before with the task

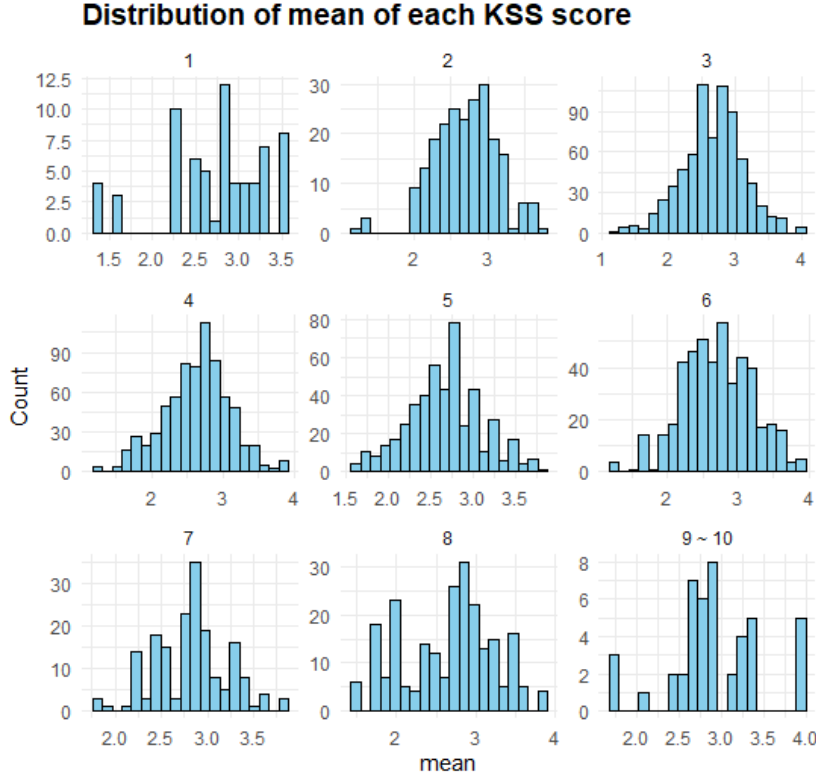


Figure 4.3: The distribution of log-transformed MEDI by KSS scores.

variable, capturing the 24-hour MEDI history as temporal dependency exists every 12 hours apart shown in Figure 4.2; the second contains the MEDI summaries and cosinor parameters with the subject and task variables. Table 4.4 shows the train and test performance scores from the Random Forest models trained under both sets of features. Models targeting reaction time performed better, with higher R^2 scores than the other metrics. The models are overfitting given the notable difference in test and train scores, where the train RMSE is doubled the test RMSE, and the train R^2 is at least 0.1 greater than the test R^2 . Similarly, the Random Forest models trained on observations from individual tasks also exhibit signs of overfitting. These models significantly perform better on the training set compared to the test set, with lower RMSE and higher R^2 values on the training data (See Table 1 and Table 2 in 'Additional Figures and Tables.pdf' in Appendix 6.1). This disparity suggests that the models do not generalize well to unseen data. Despite potential issues with the reliability of feature importance due to overfitting, notable patterns emerge across the models trained on different metrics. The subject variable consistently ranks highest in feature impor-

tance, suggesting that individual differences play a substantial role in the model's predictions. Additionally, the amplitude of the cosinor fit frequently appears as the second most important feature. This indicates that variations in MEDI exposure, both low and high, might be associated with task performance.

Table 4.2: Linear Mixed Models fitted on median reaction time, accuracy, false alarm rate, IES, and a task-specific metric (lapse, D', slope) for each task with time awake, ambient light, and their interaction term as fixed effects and the Wald test results. The p-value is bold if it is less than 0.01 significance level.

(a) PVT

Target		medianrt	lapse	ACC	FAR	ies
(Intercept)		426.48	6.26	95.25	3.42	476.92
Time awake	Coef	0.71	0.04	-0.16	0.04	2.37
	Std Error	0.472	0.038	0.063	0.039	0.949
	p-value	1.33×10^{-1}	2.65×10^{-1}	1.03×10^{-2}	2.79×10^{-1}	1.29×10^{-2}
Ambient light	Coef	-9.14	-0.38	-0.64	-0.30	-2.23
	Std Error	3.217	0.262	0.427	0.267	6.472
	p-value	4.57×10^{-3}	1.52×10^{-1}	1.35×10^{-1}	2.59×10^{-1}	7.30×10^{-1}
Time awake × Ambient light	Coef	0.17	-0.01	0.05	-0.01	-0.51
	Std Error	0.367	0.030	0.049	0.030	0.739
	p-value	6.45×10^{-1}	8.60×10^{-1}	3.17×10^{-1}	6.97×10^{-1}	4.88×10^{-1}

(b) 2-back

Target		medianrt	D'	ACC	FAR	ies
(Intercept)		646.48	2.86	94.91	3.67	740.78
Time awake	Coef	1.33	-0.01	-0.09	0.06	2.31
	Std Error	1.418	0.005	0.058	0.046	1.764
	p-value	3.50×10^{-1}	2.22×10^{-1}	1.40×10^{-1}	2.04×10^{-1}	1.91×10^{-1}
Ambient light	Coef	-15.81	-0.05	-0.58	0.34	-11.29
	Std Error	9.836	0.037	0.406	0.321	12.249
	p-value	1.08×10^{-1}	1.41×10^{-1}	1.54×10^{-1}	2.92×10^{-1}	3.57×10^{-1}
Time awake × Ambient light	Coef	-0.60	0.01	0.05	-0.02	-1.37
	Std Error	1.122	0.004	0.046	0.036	1.395
	p-value	5.94×10^{-1}	1.89×10^{-1}	2.73×10^{-1}	5.61×10^{-1}	3.26×10^{-1}

(c) VisualSearch

Target		medianrt	slope	ACC	FAR	ies
(Intercept)		2639.45	686.02	90.19	1.24	3017.94
Time awake	Coef	-1.46	3.06	-0.17	0.02	5.41
	Std Error	4.603	4.377	0.062	0.023	5.549
	p-value	7.51×10^{-1}	4.84×10^{-1}	6.36×10^{-3}	5.02×10^{-1}	3.30×10^{-1}
Ambient light	Coef	-54.53	24.23	-0.46	-0.14	-34.34
	Std Error	31.528	29.838	0.424	0.161	38.006
	p-value	8.40×10^{-2}	4.17×10^{-1}	2.77×10^{-1}	3.81×10^{-1}	3.66×10^{-1}
Time awake × Ambient light	Coef	-1.89	-4.31	-0.04	0.02	-1.03
	Std Error	3.681	3.515	0.050	0.019	4.438
	p-value	6.07×10^{-1}	2.20×10^{-1}	3.95×10^{-1}	2.92×10^{-1}	8.17×10^{-1}

Table 4.3: Linear Mixed Models fitted on median reaction time, accuracy, false alarm rate, IES, and a task-specific metric (lapse, D', slope) for each task with cosine and sine as fixed effects and the Wald test result. The p-value is bold if it is less than 0.01 significance level.

(a) PVT

Target		medianrt	lapse	ACC	FAR	ies
(Intercept)		422.60	6.15	93.61	3.30	489.10
cosine	Coef	8.23	0.38	-0.42	0.53	12.89
	Std Error	1.978	0.160	0.260	0.162	3.948
	p-value	3.47×10^{-5}	1.71×10^{-2}	1.03×10^{-1}	1.13×10^{-3}	1.14×10^{-3}
sine	Coef	2.40	0.33	0.29	0.01	0.50
	Std Error	2.689	0.217	0.354	0.220	5.365
	p-value	3.73×10^{-1}	1.29×10^{-1}	4.08×10^{-1}	9.49×10^{-1}	9.26×10^{-1}

(b) 2-back

Target		medianrt	D'	ACC	FAR	ies
(Intercept)		634.75	2.78	93.91	4.35	738.15
cosine	Coef	17.98	0.02	0.16	-0.07	19.48
	Std Error	5.957	0.022	0.244	0.193	7.396
	p-value	2.62×10^{-3}	3.07×10^{-1}	5.21×10^{-1}	7.13×10^{-1}	8.57×10^{-3}
sine	Coef	6.83	-0.02	-0.11	-0.10	15.76
	Std Error	8.092	0.030	0.332	0.262	10.047
	p-value	3.99×10^{-1}	5.56×10^{-1}	7.43×10^{-1}	7.12×10^{-1}	1.17×10^{-1}

(c) VisualSearch

Target		medianrt	slope	ACC	FAR	ies
(Intercept)		2558.81	706.49	88.02	1.37	3024.83
cosine	Coef	41.17	4.19	-0.53	0.24	69.83
	Std Error	19.411	18.502	0.263	0.099	23.300
	p-value	3.42×10^{-2}	8.21×10^{-1}	4.49×10^{-2}	1.48×10^{-2}	2.80×10^{-3}
sine	Coef	47.94	1.77	0.40	0.04	47.87
	Std Error	26.538	25.299	0.360	0.135	31.854
	p-value	7.12×10^{-2}	9.44×10^{-1}	2.62×10^{-1}	7.54×10^{-1}	1.33×10^{-1}

Table 4.4: The performance metrics of Random Forest model train trained using all observations across tasks. The features used for each Trial are stated in each result table.

(a) Random Forests trained with the MEDI history hourly mean up to 11 hours before, and the task variable.

	RMSE		R ²	
target	test	train	test	train
TPR	13.76	6.28	0.23	0.90
FDR	14.09	6.43	0.34	0.91
ACC	9.65	4.38	0.23	0.92
medianrt	412.27	173.31	0.81	0.97
meanrt	384.47	172.63	0.85	0.97
slow10	629.21	279.03	0.84	0.97
fast10	202.18	98.70	0.75	0.95

(b) Random Forest models trained with the statistical summaries and cosinor parameters of the past 24 hour MEDI observations, subject and task.

	RMSE		R ²	
target	test	train	test	train
TPR	12.88	6.00	0.32	0.90
FDR	11.84	5.54	0.53	0.93
ACC	8.59	3.96	0.39	0.91
medianrt	352.13	146.95	0.86	0.98
meanrt	329.00	144.89	0.89	0.98
slow10	507.06	244.19	0.89	0.98
fast10	178.38	92.30	0.81	0.95

Chapter 5

Discussion and Limitation

The findings in Section 4.3.1 align with Gardesevic et al. [2022], showing that subjective sleepiness significantly impacts cognitive performance, especially reaction time. In both studies, the PVT task demonstrates significant effects of subjective sleepiness on cognitive performance metrics such as median reaction time, lapses, and IES. However, there is a discrepancy regarding the impact of sleepiness on the false alarm rate in the PVT, which was not found in Gardesevic et al. [2022]. While time awake and ambient light had minimal effects, longer wakefulness did slow reaction times in the PVT, reflecting similar sleepiness effects observed in KSS models. Daily variation results generally agreed with prior research, although there are indications of weaker but significant effects of sleepiness, time awake, and daily variation on metrics in the 2-back and visual search tasks. Discrepancies in how these factors influence performance, particularly in the visual search task, could be attributed to variations in task design, sample size, and subjects compared to Gardesevic et al. [2022].

Analyzing the relationship between MEDI history and sleepiness with the ANOVA was problematic due to violations of normality and variance assumptions, making the results unreliable. Even after aggregating KSS scores, MEDI means showed minimal variance, indicating the data may not capture the relationship between MEDI history and sleepiness possibly due to limited sample size.

Random Forest models predicting cognitive performance based on MEDI history suffered from overfitting, particularly for reaction time metrics. The high importance of the subject variable suggests significant individual differences, but overfitting remains a concern.

The small, uneven sample size likely constrained the analysis, especially for MEDI history across KSS levels. A larger, more representative sample could enhance the robustness of findings. However, extending the study

could lead to participant fatigue and increased sleepiness, potentially skewing results (Mavjee and Horne [1994]). Balancing comprehensive data collection with participant well-being is crucial.

Chapter 6

Conclusion

In conclusion, this study provides valuable insights into the predictive power of various models and the impact of experimental design on the results. The findings from the linear mixed models align closely with those reported by Gardesevic et al. [2022], showing similar effects of sleepiness and day-night cycles on cognitive performance despite differences in experimental design.

ANOVA is inapplicable due to assumption violations, and Random Forest models showed overfitting with MEDI features. Subject and amplitude were the most predictive factors for cognitive performance. To improve future studies' reliability, it's recommended to collect a broader range of cognitive task records and more diverse KSS scores, enabling effective use of both ANOVA and Random Forest models.

Furthermore, for a more detailed analysis of MEDI history, Fourier analysis could be an alternative to cosinor analysis, offering higher granularity in data extraction.

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Appendix

6.1 GitHub repository

The following GitHub repository contains the working data, RScripts for the analyses. <https://github.com/edwinciw/Light-History-and-Human-Behaviour/>

6.2 Pearson Correlation Coefficient:

Pearson correlation coefficient is given by

$$\tau = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}},$$

where x_i, y_i are the i -th observation and $\bar{x}, \bar{y}$ are the mean of the variables $\mathbf{x}$ and $\mathbf{y}$, respectively, for $i \in \{1, 2, \dots, n\}$. The coefficient τ ranges from -1 to 1 , the greater the value of the absolute value $|\tau|$, the stronger the linear relationship between $\mathbf{x}$ and $\mathbf{y}$. The sign of τ indicates negative linear relationship if $-$, else positive for $+$.

6.3 One-way ANOVA

One-way ANOVA is based on the model

$$Y_{ij} = \mu + a_i + \epsilon_{ij}, \quad (6.1)$$

where μ is the mean of a numeric variable $\mathbf{Y}$ of light history, a_i is the fixed effect of score i from $\mathbf{G}$ indicating the score effect of score i in $\mathbf{Y}$, Y_{ij} is the j -th observation in the partition $\mathbf{Y}_i$ of $\mathbf{Y}$ from score i , and $\epsilon_{ij} \sim N(0, \sigma^2)$ simulates random error normally distributed with fixed variance $\sigma^2 < \infty$, for all $i \in \{1, 2, \dots, k\}$ and $j \in \{1, 2, \dots, N\}$. Overall, $Y_{ij} \sim N(\mu + a_i, \sigma^2)$ for each i, j suggesting the partition of each score i is normally distributed for an adequate fit with the model 6.1.

6.4 F-statistic

The F-statistic is given by

$$F = \frac{\text{Mean square between groups}}{\text{Mean square within groups}} = \frac{\frac{1}{k-1} \sum_{i=1}^k n_i (\bar{Y}_i - \mu)^2}{\frac{1}{N-k} \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2}, \quad (6.2)$$

with $\bar{Y}_i$ as the mean of $\mathbf{Y}_i$, and $F \sim \frac{\chi_{k-1}^2 / (k-1)}{\chi_{N-k}^2 / (N-k)}$ is a $F_{k-1, N-k}$ -distribution with chi-square distributions χ_1^2 / σ^2 and χ_2^2 / σ^2 are sum of independent standard normal variables with degrees of freedom df as $k-1$ and $N-k$, respectively (Kendall and Stuart [1979]). Continue with the hypothesis test, for a significance level α , find the critical value $F_{\alpha, k-1, N-k}$ such that

$$P(F > F_{\alpha, k-1, N-k} | H_0) = \alpha,$$

where the area under the curve of $F_{k-1, N-k}$ equal to α between $F_{\alpha, k-1, N-k}$ to the right of the distribution. If the F -statistic 6.2 estimated with $\mathbf{K}$ and $\mathbf{Y}$ is greater than the critical value $F_{\alpha, k-1, N-k}$, then the mean square between groups is significantly greater than within groups,

6.5 Wald Test

The Wald statistic is calculated as:

$$W_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)},$$

where $\hat{\beta}_j$ is the estimated coefficient of the j -th fixed effect from β , and $SE(\hat{\beta}_j)$ is its standard error. The Wald test assesses the significance of the correlation between the fixed effects and the response variable under the null hypothesis that the coefficients are zero, with W_j following a standard normal distribution $N(0, 1)$ if the null hypothesis is true.

In the context of mixed models, the distribution of the Wald statistic W_j is often better approximated by a t-distribution rather than a normal distribution due to the lack of samples per subject, with the degrees of freedom for the t-distribution are typically adjusted using methods such as the Satterthwaite approximation (Kuznetsova et al. [2017], Satterthwaite [1946]), which accounts for the uncertainty introduced by the random effects.

The significance of the fixed effects is then assessed by computing the p-value, which is defined as:

$$p\text{-value} = 2 \times (1 - \text{CDF}_{t_{df}}(|W_j|)),$$

where $\text{CDF}_{t_{\text{df}}}$ is the cumulative distribution function of the t-distribution with the appropriate degrees of freedom df . A small p-value (typically less than 0.01) indicates significant evidence to reject null hypothesis.

6.6 RMSE and R^2

The performance metrics RMSE and R^2 are given by

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2},$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2},$$

where y_i is the actual target, $\hat{y}_i$ is the predicted target, $\bar{y}$ is the mean of the actual values, and n is the number of observations.